On the State ψ_{CROWN} and Its Stability Under Diagonal Unitary Evolution

Abstract

We introduce a normalized equal-modulus pure state

$$|\psi_{\text{CROWN}}\rangle = \frac{1}{\sqrt{n+1}} \sum_{k=0}^{n} e^{i\theta_k} |k\rangle$$

in a finite-dimensional Hilbert space and study its behavior under unitary dynamics generated by a diagonal self-adjoint Hamiltonian. Two notions of stability are distinguished: invariance of measurement probabilities in the reference basis, and invariance of relative phases between basis amplitudes. We prove that basis probabilities are preserved under arbitrary diagonal unitary evolution, while exact phase-pattern preservation requires a degeneracy condition on the relevant part of the spectrum. The presentation is fully explicit and self-contained.

1 Introduction

Let $\mathcal{H}$ be a complex Hilbert space of finite dimension $n+1$, equipped with a fixed orthonormal basis

$$\mathcal{B} = \{|0\rangle, |1\rangle, \ldots, |n\rangle\}.$$

We consider a family of normalized pure states with equal amplitude magnitudes and arbitrary initial phase profile. Such states are simple enough to admit complete analysis, while still exhibiting a nontrivial distinction between two kinds of invariance under time evolution:

(i) invariance of the probability distribution associated with the basis $\mathcal{B}$,

(ii) invariance of the relative phase structure of the state.

These two notions are often conflated informally. In fact, they are mathematically distinct. The first is automatic under diagonal unitary evolution, whereas the second generally fails unless additional spectral constraints are imposed.

The purpose of this note is to formalize this distinction and to provide complete proofs.

2 Preliminaries

We work throughout in a finite-dimensional Hilbert space $\mathcal{H} \cong \mathbb{C}^{n+1}$. The inner product is denoted by $\langle \cdot, \cdot \rangle$, linear in the second argument and conjugate-linear in the first. The basis $\mathcal{B} = \{|k\rangle\}_{k=0}^{n}$ is orthonormal:

$$\langle j \mid k \rangle = \delta_{jk}.$$

A vector $|\psi\rangle \in \mathcal{H}$ is normalized if

$$\langle \psi \mid \psi \rangle = 1.$$

Let $H : \mathcal{H} \to \mathcal{H}$ be a self-adjoint operator. The associated unitary time evolution is

$$U(t) = e^{-itH}, \qquad t \in \mathbb{R}.$$

If H is diagonal in the basis $\mathcal{B}$, then there exist real numbers $\lambda_0, \ldots, \lambda_n$ such that

$$H|k\rangle = \lambda_k |k\rangle \qquad \text{for all } k = 0, \ldots, n.$$

Consequently,

$$U(t)|k\rangle = e^{-it\lambda_k}|k\rangle.$$

3 Definition of the state ψ_{CROWN}

Definition 3.1 (ψ_{CROWN})**.** *Let $\theta_0, \ldots, \theta_n \in \mathbb{R}$. Define*

$$|\psi_{\text{CROWN}}\rangle = \frac{1}{\sqrt{Z}} \sum_{k=0}^{n} e^{i\theta_k}|k\rangle, \tag{3.1}$$

where $Z > 0$ is chosen so that $|\psi_{\text{CROWN}}\rangle$ is normalized.

Lemma 3.2 (Normalization)**.** *For the state defined in* (3.1)*, the normalization constant is*

$$Z = n + 1.$$

Proof. Using orthonormality of the basis,

$$\begin{aligned}
\langle \psi_{\text{CROWN}} \mid \psi_{\text{CROWN}} \rangle &= \left(\frac{1}{\sqrt{Z}} \sum_{j=0}^{n} e^{-i\theta_j} \langle j| \right) \left(\frac{1}{\sqrt{Z}} \sum_{k=0}^{n} e^{i\theta_k}|k\rangle \right) \\
&= \frac{1}{Z} \sum_{j=0}^{n} \sum_{k=0}^{n} e^{i(\theta_k - \theta_j)} \langle j \mid k \rangle \\
&= \frac{1}{Z} \sum_{j=0}^{n} \sum_{k=0}^{n} e^{i(\theta_k - \theta_j)} \delta_{jk} \\
&= \frac{1}{Z} \sum_{k=0}^{n} 1 = \frac{n+1}{Z}.
\end{aligned}$$

Normalization requires $\langle \psi_{\text{CROWN}} \mid \psi_{\text{CROWN}} \rangle = 1$, hence

$$\frac{n+1}{Z} = 1,$$

which implies $Z = n + 1$. □

Therefore the state can be written explicitly as

$$|\psi_{\text{CROWN}}\rangle = \frac{1}{\sqrt{n+1}} \sum_{k=0}^{n} e^{i\theta_k}|k\rangle. \tag{3.2}$$

Remark 3.3. Each basis component has the same modulus $1/\sqrt{n+1}$, while the phase profile $(\theta_0, \ldots, \theta_n)$ is arbitrary.

4 Diagonal unitary evolution

We now compute the exact form of the time-evolved state under a diagonal Hamiltonian.

Lemma 4.1 (Explicit evolution formula)**.** *Let H be diagonal in the basis $\mathcal{B}$, with*

$$H|k\rangle = \lambda_k|k\rangle, \qquad \lambda_k \in \mathbb{R}.$$

Then, for all $t \in \mathbb{R}$,

$$U(t)|\psi_{\text{CROWN}}\rangle = \frac{1}{\sqrt{n+1}} \sum_{k=0}^{n} e^{i(\theta_k - \lambda_k t)}|k\rangle. \tag{4.1}$$

Proof. Since $|k\rangle$ is an eigenvector of H with eigenvalue λ_k, it is also an eigenvector of $U(t) = e^{-itH}$ with eigenvalue $e^{-it\lambda_k}$. Thus

$$U(t)|k\rangle = e^{-it\lambda_k}|k\rangle.$$

By linearity and (3.2),

$$\begin{aligned} U(t)|\psi_{\text{CROWN}}\rangle &= U(t)\left(\frac{1}{\sqrt{n+1}} \sum_{k=0}^{n} e^{i\theta_k}|k\rangle\right) \\ &= \frac{1}{\sqrt{n+1}} \sum_{k=0}^{n} e^{i\theta_k} U(t)|k\rangle \\ &= \frac{1}{\sqrt{n+1}} \sum_{k=0}^{n} e^{i\theta_k} e^{-it\lambda_k}|k\rangle \\ &= \frac{1}{\sqrt{n+1}} \sum_{k=0}^{n} e^{i(\theta_k - \lambda_k t)}|k\rangle. \end{aligned}$$

□

5 Probability stability in the reference basis

We first establish the easiest and most robust form of stability: the invariance of basis probabilities.

Theorem 5.1 (Basis-probability invariance)**.** *Let H be diagonal in the basis $\mathcal{B}$, and let $U(t) = e^{-itH}$. Then for every $k \in \{0, \ldots, n\}$ and every $t \in \mathbb{R}$,*

$$|\langle k|U(t)|\psi_{\text{CROWN}}\rangle|^2 = |\langle k|\psi_{\text{CROWN}}\rangle|^2 = \frac{1}{n+1}. \tag{5.1}$$

Proof. From Lemma 4.1,

$$U(t)|\psi_{\text{CROWN}}\rangle = \frac{1}{\sqrt{n+1}} \sum_{j=0}^{n} e^{i(\theta_j - \lambda_j t)}|j\rangle.$$

Taking the inner product with $\langle k|$, we obtain

$$\begin{aligned} \langle k|U(t)|\psi_{\text{CROWN}}\rangle &= \frac{1}{\sqrt{n+1}} \sum_{j=0}^{n} e^{i(\theta_j - \lambda_j t)}\langle k \mid j\rangle \\ &= \frac{1}{\sqrt{n+1}} \sum_{j=0}^{n} e^{i(\theta_j - \lambda_j t)}\delta_{kj} \\ &= \frac{1}{\sqrt{n+1}} e^{i(\theta_k - \lambda_k t)}. \end{aligned}$$

Hence

$$|\langle k|U(t)|\psi_{\text{CROWN}}\rangle|^2 = \frac{1}{n+1}\left|e^{i(\theta_k-\lambda_k t)}\right|^2.$$

Since $|e^{i\varphi}| = 1$ for every real φ, it follows that

$$|\langle k|U(t)|\psi_{\text{CROWN}}\rangle|^2 = \frac{1}{n+1}.$$

This proves the statement. □

Corollary 5.2. *Under arbitrary diagonal unitary evolution, the state $|\psi_{\text{CROWN}}\rangle$ remains uniformly distributed in the basis $\mathcal{B}$.*

Proof. Immediate from Theorem 5.1. □

Remark 5.3. This result does not require any degeneracy assumption on the spectrum of H. The reason is simple: diagonal evolution changes only the phase of each basis amplitude and does not alter its modulus.

6 Relative phase stability

We now turn to the stronger question of whether the *relative phases* of the amplitudes are preserved.

Definition 6.1 (Relative phase difference)**.** *Let*

$$|\psi\rangle = \sum_{k=0}^{n} c_k|k\rangle$$

with $c_k \neq 0$. For indices j, k, define the relative phase difference by

$$\Delta_{kj}(\psi) = \arg(c_k) - \arg(c_j) \quad \textit{mod } 2\pi.$$

For the state $|\psi_{\text{CROWN}}\rangle$, the initial relative phase difference is

$$\Delta_{kj}(0) = \theta_k - \theta_j \quad \text{mod } 2\pi.$$

After time evolution, by (4.1), the coefficient of $|k\rangle$ becomes

$$\frac{1}{\sqrt{n+1}}e^{i(\theta_k-\lambda_k t)},$$

so the evolved relative phase difference is

$$\Delta_{kj}(t) = (\theta_k - \lambda_k t) - (\theta_j - \lambda_j t) = (\theta_k - \theta_j) - (\lambda_k - \lambda_j)t \quad \text{mod } 2\pi. \tag{6.1}$$

Equation (6.1) shows immediately that relative phases are generally time-dependent unless the spectral gaps vanish in the relevant directions.

Theorem 6.2 (Characterization of exact relative-phase stability)**.** *Let H be diagonal in the basis $\mathcal{B}$, with eigenvalues $\lambda_0, \ldots, \lambda_n$. Then the following are equivalent:*

(i) for all $t \in \mathbb{R}$ and all $j, k \in \{0, \ldots, n\}$,

$$\Delta_{kj}(t) = \Delta_{kj}(0) \quad \textit{mod } 2\pi;$$

(ii) for all $j, k \in \{0, \ldots, n\}$,

$$\lambda_k = \lambda_j;$$

(iii) there exists $\lambda \in \mathbb{R}$ such that

$$H|k\rangle = \lambda|k\rangle \quad \textit{for all } k = 0, \ldots, n.$$

In that case,

$$U(t)|\psi_{\text{CROWN}}\rangle = e^{-i\lambda t}|\psi_{\text{CROWN}}\rangle \qquad \textit{for all } t \in \mathbb{R}. \tag{6.2}$$

Proof. We prove $(i) \Rightarrow (ii) \Rightarrow (iii) \Rightarrow (i)$.
$(i) \Rightarrow (ii)$. Assume that for all $t \in \mathbb{R}$ and all j, k,

$$\Delta_{kj}(t) = \Delta_{kj}(0) \mod 2\pi.$$

Using (6.1), this means

$$(\theta_k - \theta_j) - (\lambda_k - \lambda_j)t = \theta_k - \theta_j \mod 2\pi$$

for all t, hence

$$(\lambda_k - \lambda_j)t = 0 \mod 2\pi$$

for all $t \in \mathbb{R}$. The only way this can hold for every real t is

$$\lambda_k - \lambda_j = 0.$$

Thus $\lambda_k = \lambda_j$ for all j, k.
$(ii) \Rightarrow (iii)$. If $\lambda_k = \lambda_j$ for all j, k, then all eigenvalues are equal to a common real number λ. Hence

$$H|k\rangle = \lambda|k\rangle$$

for all k.
$(iii) \Rightarrow (i)$. Assume there exists $\lambda \in \mathbb{R}$ such that

$$H|k\rangle = \lambda|k\rangle$$

for every k. Then

$$U(t)|k\rangle = e^{-i\lambda t}|k\rangle,$$

so

$$\begin{aligned} U(t)|\psi_{\text{CROWN}}\rangle &= \frac{1}{\sqrt{n+1}} \sum_{k=0}^{n} e^{i\theta_k} U(t)|k\rangle \\ &= \frac{1}{\sqrt{n+1}} \sum_{k=0}^{n} e^{i\theta_k} e^{-i\lambda t}|k\rangle \\ &= e^{-i\lambda t} \frac{1}{\sqrt{n+1}} \sum_{k=0}^{n} e^{i\theta_k}|k\rangle \\ &= e^{-i\lambda t}|\psi_{\text{CROWN}}\rangle. \end{aligned}$$

Thus the state changes only by a global phase. Global phase multiplication leaves all relative phases unchanged, so

$$\Delta_{kj}(t) = \Delta_{kj}(0) \mod 2\pi$$

for all j, k and t.

This proves the equivalence as well as (6.2). □

Corollary 6.3. *Exact preservation of the full phase pattern is strictly stronger than preservation of basis probabilities.*

Proof. By Theorem 5.1, basis probabilities are preserved for every diagonal Hamiltonian. By Theorem 6.2, exact preservation of relative phases requires complete spectral degeneracy on the support of the state. Therefore the second property is stronger. □

7 Interpretation

The results obtained above can be summarized as follows.

Weak stability

Under arbitrary diagonal unitary evolution, the state $|\psi_{\text{CROWN}}\rangle$ preserves its uniform basis probability distribution. In concrete terms,

$$|\langle k|U(t)|\psi_{\text{CROWN}}\rangle|^2 = \frac{1}{n+1}$$

for every basis vector and every time t.

Strong stability

If one asks for preservation of the entire phase profile up to a physically irrelevant global phase, then diagonality alone is not enough. One must require that the diagonal generator acts as a scalar on the span of the basis components appearing in the state. In the present setup, since every basis vector appears with nonzero amplitude, this means

$$H = \lambda I$$

on the whole space $\mathcal{H}$, or at least on the support of the state.

Remark 7.1. If one considers a more general equal-modulus state supported only on a subset $S \subseteq \{0, \ldots, n\}$, then exact phase-pattern invariance requires only that the eigenvalues λ_k be equal for $k \in S$. The proof is identical.

8 Generalization to non-uniform amplitudes

For completeness, we note that the same distinction persists for arbitrary nonzero amplitudes.

Let

$$|\phi\rangle = \sum_{k=0}^{n} a_k e^{i\theta_k}|k\rangle, \qquad a_k \geq 0, \qquad \sum_{k=0}^{n} a_k^2 = 1.$$

Under diagonal evolution,

$$U(t)|\phi\rangle = \sum_{k=0}^{n} a_k e^{i(\theta_k - \lambda_k t)}|k\rangle.$$

Then:

(i) the basis probabilities remain invariant:

$$|\langle k|U(t)|\phi\rangle|^2 = a_k^2;$$

(ii) exact preservation of all relative phases among indices with $a_k \neq 0$ holds if and only if all corresponding eigenvalues λ_k are equal.

We omit the proof, since it is a straightforward repetition of the arguments already given.

9 Conclusion

We introduced the state

$$|\psi_{\text{CROWN}}\rangle = \frac{1}{\sqrt{n+1}} \sum_{k=0}^{n} e^{i\theta_k} |k\rangle$$

and analyzed its time evolution under diagonal self-adjoint generators.

The main conclusions are:

(1) the state is normalized by the factor $1/\sqrt{n+1}$;

(2) under arbitrary diagonal unitary evolution, the probability of measuring any basis state $|k\rangle$ remains equal to $1/(n+1)$;

(3) exact preservation of the relative phase pattern occurs if and only if the diagonal generator is spectrally degenerate on the support of the state;

(4) in the fully degenerate case, the evolution reduces to multiplication by a global phase.

Thus the mathematically correct statement is not simply that the state is "stable", but rather that it exhibits *probability stability* under diagonal evolution and *phase rigidity* precisely under spectral degeneracy.

This closes the argument.